Module 1

Individual Task

Research and present at timeline showing major milestone in AI history

Introduction:

Artificial Intelligence (AI) is the field of computer science focused on creating systems capable of performing tasks that normally require human intelligence, such as reasoning, learning, perception, and language understanding.

Since its formal birth in the mid-20th century, AI has progressed through cycles of optimism, disappointment, and breakthrough innovation.

Today, AI powers search engines, medical diagnostics, autonomous vehicles, and conversational systems. Understanding its history helps explain current capabilities and future directions.

Foundations of AI (1940s–1960s):

1943 — First Mathematical Model of Neural Networks

Warren McCulloch and Walter Pitts created the first mathematical model of artificial neurons. This work demonstrated that machines could theoretically simulate human brain processes.

Impact:

- Established theoretical basis for neural networks
- Inspired future machine learning research

1950 — The Turing Test

Alan Turing published 'Computing Machinery and Intelligence', proposing the famous **Turing Test** to evaluate machine intelligence.

Impact:

- Provided philosophical foundation for AI
- Shifted focus to observable intelligent behavior.

1956 — Dartmouth Conference (Birth of AI)

The Dartmouth Summer Research Project formally established AI as a scientific field.

Key figures:

- John McCarthy
- Marvin Minsky
- Claude Shannon
- Nathaniel Rochester

Impact:

- Coined the term “Artificial Intelligence”
- Launched organized AI research

Growth and AI Winters (1970s–1990s)

1974–1980 — First AI Winter:

Funding declined after early AI systems failed to meet expectations.

Causes:

- Over-promising by researchers
- Limited computing power
- Poor real-world performance

Impact:

- Major reduction in research funding
- Slowed progress for nearly a decade

1980 — Rise of Expert Systems:

Expert systems like XCON became commercially successful.

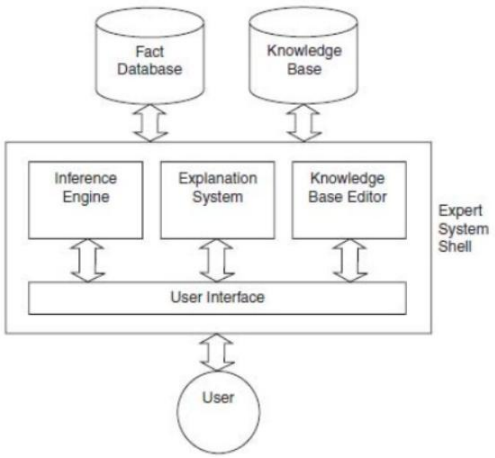


Fig Expert System Architecture

Impact:

- First profitable AI applications
- Heavy corporate investment
- Widespread adoption in industry

1987–1993 — Second AI Winter:

Another collapse occurred due to expensive and brittle expert systems.

Impact:

- Funding cuts
- Shift toward machine learning approaches

1997 — Deep Blue Defeats Kasparov:

IBM's Deep Blue beat world chess champion Garry Kasparov.

Impact:

- Major public milestone
- Demonstrated brute-force AI power
- Renewed interest in AI

The Modern AI Revolution (2000–2015)

2006 — Deep Learning Revival

Geoffrey Hinton and others reignited neural network research.

Impact:

- Shift from symbolic AI to data-driven learning
- Beginning of modern AI boom

2012 — AlexNet Wins ImageNet

A deep neural network dramatically improved image recognition accuracy.

Impact:

- Proved superiority of deep learning
- Massive investment in GPUs
- Sparked AI race among tech companies

2016 — AlphaGo Defeats Lee Sedol

Google DeepMind’s AlphaGo beat a world Go champion.

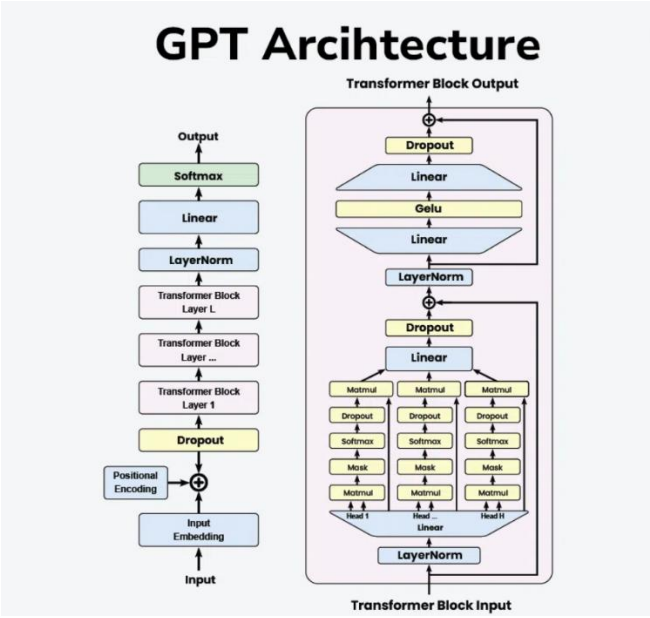
Impact:

- Showed AI mastery of complex intuition-based tasks
- Major milestone in reinforcement learning
- Global media attention

Generative AI Era (2018–2023)

2020 — GPT-3 Released

OpenAI introduced GPT-3, a powerful large language model.



Impact:

- Demonstrated few-shot learning
- Enabled advanced text generation
- Accelerated commercial AI adoption

2020 — AlphaFold2 Solves Protein Folding

DeepMind’s AlphaFold2 achieved unprecedented accuracy in protein prediction.

Impact:

- Solved a 50-year biology challenge

- Transformed drug discovery
- Showed AI's scientific power

2022 — ChatGPT Launch

ChatGPT brought conversational AI to the mainstream, reaching massive adoption quickly.

Impact:

- Fastest-growing consumer app
- Triggered global AI race
- Popularized generative AI

2023 — AI Reaches 100 Million Users

ChatGPT surpassed 100 million users within months.

Impact:

- Demonstrated mass adoption
- Shifted public perception of AI
- Accelerated regulation efforts

Recent Developments and Future Outlook

2024 — Global AI Regulation Begins

The European Union formally adopted the AI Act in 2024, one of the first comprehensive AI laws.

Impact:

- Beginning of global AI governance
- Focus on safety and transparency
- Model for other countries

2024 — Nobel Recognition for Neural Networks

Geoffrey Hinton and John Hopfield received the Nobel Prize in Physics for neural-network research.

Impact:

- Validated AI as fundamental science
- Highlighted deep learning importance
- Raised awareness of AI risks

Future Trends

AI is expected to advance in:

- Artificial general intelligence (AGI)
- Multimodal AI systems
- Autonomous agents
- AI regulation and ethics
- Human-AI collaboration

Researchers note that AI innovation is accelerating rapidly, with research output and participation growing exponentially.

Conclusion

Artificial Intelligence has evolved from theoretical speculation in the 1940s to one of the most transformative technologies of the 21st century.

The field has experienced cycles of hype and disappointment, but breakthroughs in deep learning and large language models have ushered in a new era.

The timeline shows a clear pattern:

- **1950s–60s:** Foundations
- **1970s–90s:** Winters and recovery
- **2000s–2010s:** Deep learning revolution
- **2020s:** Generative AI explosion

As computing power, data availability, and algorithms continue to improve, AI will likely become even more deeply embedded in society.

Understanding its history is essential for guiding its responsible future development.